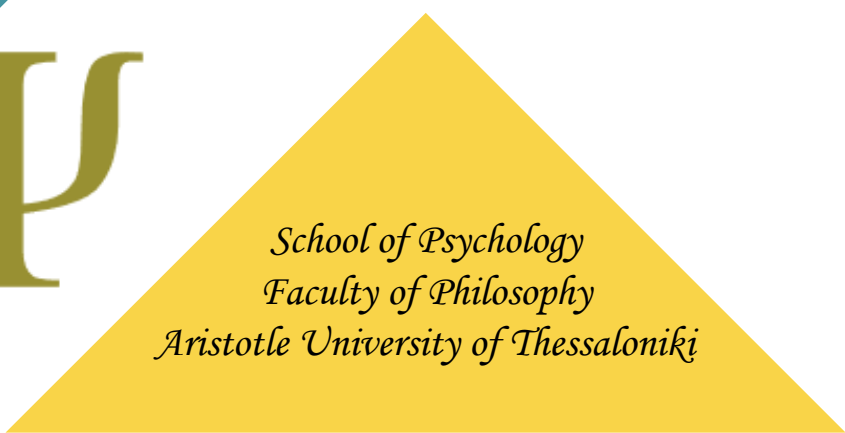




Two-way Analysis of Variance (ANOVA)

Between-subjects designs

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Έλεγχος υποθέσεων-Βήματα

1. Καθορίζεται η **μηδενική υπόθεση** H_0 (=) και **εναλλακτική υπόθεση** H_1 (≠) .
2. Ορίζεται το **επίπεδο σημαντικότητας** α (συνήθως $\alpha=0.05$).
3. Επιλέγεται μια κατάλληλη **στατιστική δοκιμασία** και υπολογίζεται η τιμή του στατιστικού με βάση τα δεδομένα του δείγματος.
4. Σύγκριση της **πιθανότητας** p να έχουμε την συγκεκριμένη τιμή του στατιστικού (ή κάτι πιο ακραίο) θεωρώντας ότι ισχύει η H_0 , με το **επίπεδο σημαντικότητας** α (0.05). Στατιστικά σημαντικό αποτέλεσμα ($p < 0.05$).
5. **Ερμηνεία** αποτελεσμάτων.

Two-way between-subjects ANOVA

ANOVA: Y = numeric variable $\Rightarrow$ **Dependent**

X_1 = categorical (factor A) variable $\Rightarrow$ **Independent**

X_2 = categorical (factor B) variable $\Rightarrow$ **Independent**

EXAMPLE

Assume that researchers are interested in exploring the role of **analogical thinking** in the **problem-solving skills** of **children and adolescents**.

Y: **Number of mistakes** made while attempting to solve the problems

X1: **intervention** control group, exp. group 1 (exposure to similar examples), and exp. group 2 (exposure + instructions)

X2: **school age** (primary school, secondary school)

Two-way between- subjects ANOVA

Null and alternative hypotheses

Main effect: intervention

- H_0 : There is no significant difference in number of mistakes between intervention groups ($\mu_{control} = \mu_{exper1} = \mu_{exper2}$)
- H_1 : At least one intervention has a significantly different mean number of mistakes.

Main effect: school age

- H_0 : There is no significant difference in number of mistakes between primary and secondary school. ($\mu_{primary} = \mu_{secondary}$)
- H_1 : There is a significant difference in number of mistakes between primary and secondary school ($\mu_{primary} \neq \mu_{secondary}$).

Interaction effect between intervention and school age

- H_0 : There is no interaction effect between intervention and school age on number of mistakes (i.e., the effect of intervention on number of mistakes does not depend on school age).
- H_1 : There is a significant interaction effect between intervention and school age on number of mistakes (i.e., the effect of intervention on number of mistakes varies depending on school age).

Assumptions of two-way between-subjects ANOVA

- Between-subjects design
- **Normality:** The number of mistakes should be approximately normally distributed within each group (i.e., for each combination of intervention and school age).
 - ➡ Check the residuals (Shapiro-Wilk test)
- **Homogeneity of Variance (Homoscedasticity):** The variance of number of mistakes should be approximately equal across all groups.
 - ➡ Check for homogeneity of variances (Levene's test)

DATA (3x2)

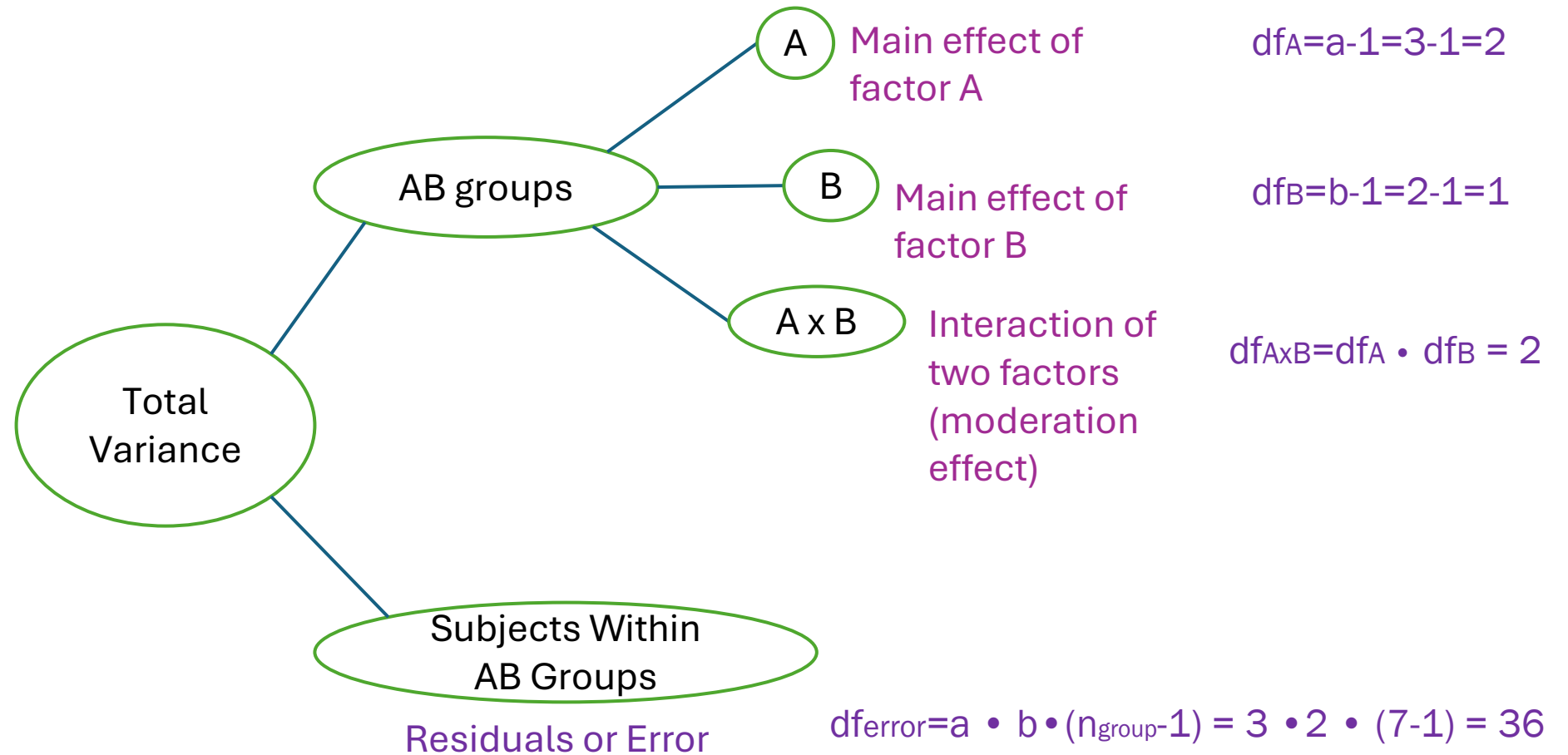
How it works

Factor B

		Factor A			MEAN
		control	exp1	exp2	
Secondary		38	20	10	
		47	20	8	
		42	22	12	
		37	24	5	
		33	32	6	
		31	19	0	
		33	36	0	
	Mean	37.3	24.7	5.9	22.6
Primary		34	29	37	
		39	47	45	
		37	28	48	
		26	43	29	
		46	41	34	
		40	44	32	
		27	31	51	
	Mean	35.6	37.6	39.4	37.5
	MEAN	36.4	31.1	22.6	30.1

Two-way between-subjects ANOVA

Partitioning the total variance into its sources



Interaction

An interaction occurs when the effect of one factor changes according to the level of the other factor.

→ E.g., the number of mistakes in the exposure-to-examples group is lower among secondary school students than among primary school students.



ANOVA TABLE (two-factors with interaction)

ANOVA Table (type II tests)

	Effect	SSn	SSd	DFn	DFd	F	p	p<.05	ges
Main effect A →	1 intervention	1354.429	1714.857	2	36	14.217	2.81e-05	*	0.441
Main effect B →	2 school_age	2332.595	1714.857	1	36	48.968	3.31e-08	*	0.576
Interaction →	3 intervention:school_age	2200.905	1714.857	2	36	23.102	3.51e-07	*	0.562

There is significant interaction between intervention and school age!

A key principle in interpreting and reporting factorial analysis results is that **interactions** take precedence over main effects.

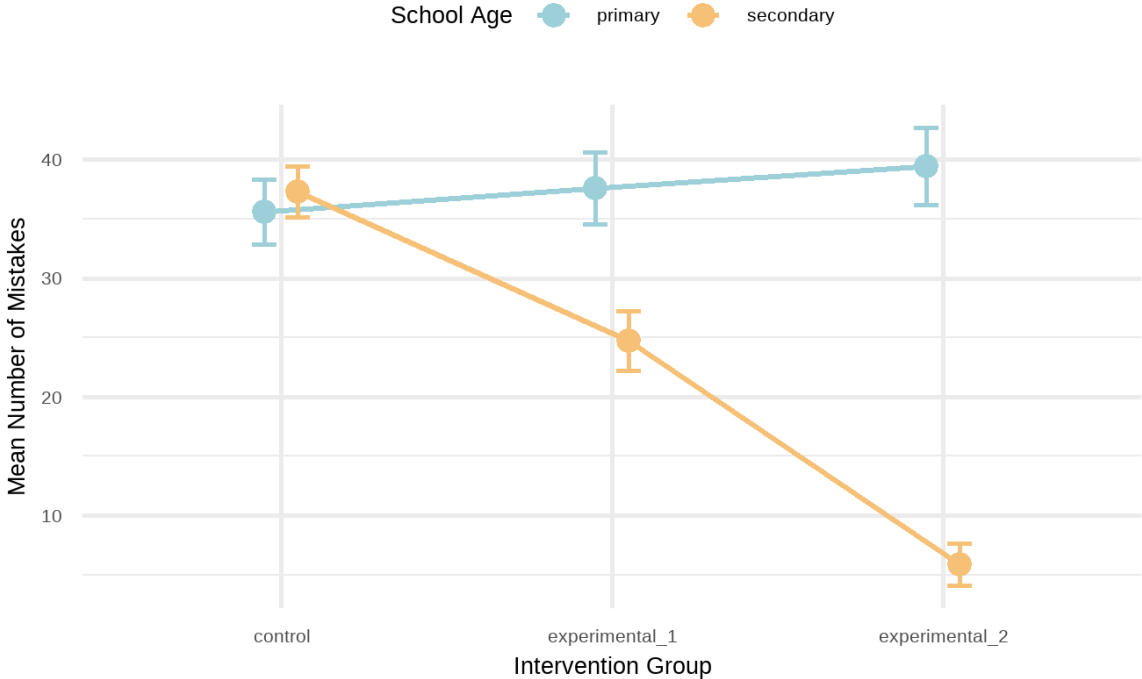
→ Conduct a **simple effects** analysis

Two-way between-subjects ANOVA

Simple effects analysis

Effect of Intervention on Mistakes by School Age

Interaction Plot with Standard Error bars



ANOVA for Simple Effects of Intervention

Moderator: School age

A tibble: 2 × 8

	school_age	Effect	DFn	DFd	F	p	`p<.05`	ges
*	<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	primary	intervention	2	36	0.547	0.584	""	0.029
2	secondary	intervention	2	36	36.8	0.000000002	"**"	0.671

ANOVA for Simple Effects of School age

Moderator: Intervention

A tibble: 3 × 8

	intervention	Effect	DFn	DFd	F	p	`p<.05`	ges
*	<fct>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>
1	control	school_age	1	36	0.216	6.45e- 1	""	0.006
2	experimental_1	school_age	1	36	12.1	1 e- 3	"**"	0.252
3	experimental_2	school_age	1	36	82.8	7.25e-11	"**"	0.697

Post hoc Tests

- Bonferroni Procedure
- Tukey's Test
- Scheffe
- Holm